01

TITLE

UNIVERSITY OF CAMERINO · FINAL PROJECT

UniBridge

A multilingual assistant that answers international students from official Italian documents, in their own language, with the Italian source of every sentence.

COURSE
Complex Systems Design

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STUDENT ASKS · ARABIC

متى آخر موعد لتقديم طلب المنحة الدراسية؟

UNIBRIDGE ANSWERS · ARABIC

الموعد النهائي هو 28 أغسطس 2026. [1]

[1] ERDIS call 2026/2027, Art. 10 · «entro il 28 agosto 2026, pena l'esclusione dal concorso»

17 LANGUAGES

it en ar fr es pt de ro sq uk ru tr fa ur hi bn zh

20 sec

Objectives and research question

Can an assistant help international students with Italian rules without inventing anything?

PROBLEM

Fees, career rules, admission and the regional scholarship are described in Italian documents. Missing a deadline excludes a student from the scholarship (Art. 10 of the call). General chatbots do not show their source and can invent dates.

RESEARCH QUESTION

How well can search over official documents, a translation step and a verified language-model answer serve questions asked in many languages, and when must the system say "I don't know"?

01

Cited answers

Each sentence ends with its passage; the Italian original next to it.

02

Every language

Detect it, search across languages, answer in it.

03

Scholarship guide

Eligibility, checklist and calendar from fixed rules.

04

Knowing the limits

Decline out-of-scope questions; e-mail to the right office.

PRIMARY METRICS

nDCG@10 · recall@10 · citation precision · answer language

DIAGNOSTIC METRICS

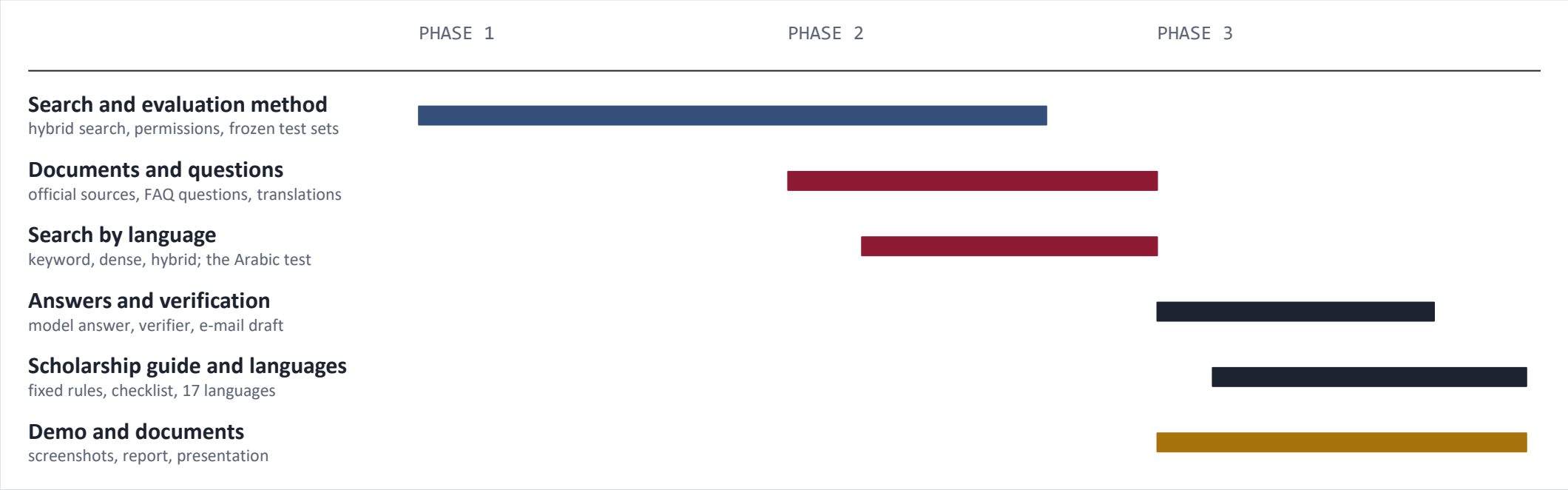
false "I don't know" · permission leaks · cost per answer · latency

55 sec

HOW THE WORK WAS ORGANISED

Three phases, one author

Each phase ends with a check that the next one depends on



Gate 1

Search tested (permissions, contracts); evaluation method reproducible.

Gate 2

Test questions frozen with a SHA-256 hash before any tuning.

Gate 3

All tests green; every number checked against the result files.

What each phase delivered

PHASE 1

Foundation

Safe parsing of PDF, Word and HTML; passages with stable ids; keyword and multilingual search with rank fusion; permissions applied inside the search; citations; a REST API with frozen contracts; the evaluation method.

PHASE 2

Data and first findings

42 official documents with licence and hash; roles for visitors, students and offices; real questions from official FAQ pages; English and Arabic versions; search measured by language. Finding: keyword search fails on Arabic.

PHASE 3

The assistant

Answers with a verifier and an extractive fallback; "I don't know" with an office e-mail; the scholarship guide; 17 languages; demo, report and presentation. Shown on Friday.

System flow and software architecture

One service · four components · an optional language model

01

Documents

Official sources parsed into passages with document, section and page. 2,330 passages.

02

Search

Keyword + multilingual embeddings, rank fusion; the user's permissions applied inside the search.

03

Answer

The model answers only from the numbered passages; a verifier drops unsupported sentences.

04

Student or office

Answer with Italian sources, or "I don't know" and an Italian e-mail draft. Nothing is sent automatically.

Without a key

No model configured: extractive answers, Italian sentences from the documents.

Passages are data

Wrapped in tags the text cannot close; the prompt says they are never instructions.

Cost control

Temperature 0, cached responses, every call logged, hard stop at 5 USD.

Components and contracts

PRESENTATION

Web demo · chat with roles and language selector · scholarship guide · e-mail draft download

SERVICE API

POST /v1/answer · POST /v1/handoff · POST /v1/procedure/scholarship · POST /v1/search · GET /v1/health

ANSWER

language detection, translation, answer, verifier, extractive fallback

HANDOFF

office rule table, Italian e-mail

SCHOLARSHIP GUIDE

fixed rules, checklist, calendar; no language model

LANGUAGE MODEL · OPTIONAL

OpenAI or Anthropic, cache, usage log, budget

SEARCH

keyword index + multilingual embeddings, rank fusion, permissions, citations

STORAGE

SQLite (documents, passages, logs) · Qdrant (vectors) · Docker

Contracts frozen as OpenAPI and JSON Schema; every response names the index and model versions. Roles: visitor (public) · student (+ student-only notices) · office staff (+ own internal notes).

FROM QUESTION TO ANSWER

A question and its constraints

One question follows the same route in every language

01

Language

Detected offline among 17 languages.

02

Search as written

Passages that share meaning or words with the question.

03

Translate and fuse

Italian translation, second search, rankings fused.

04

Answer and verify

One claim per sentence, each with [Pn]; unsupported sentences removed.

05

Show

Answer in the student's language next to the Italian original.

Weak evidence

No passage passes the threshold: "I don't know" in the student's language + e-mail draft.

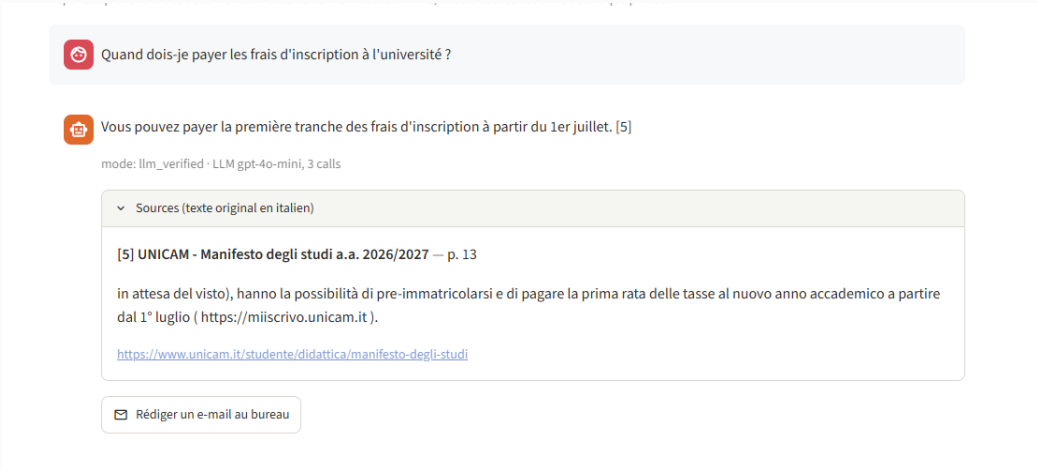
Out of scope

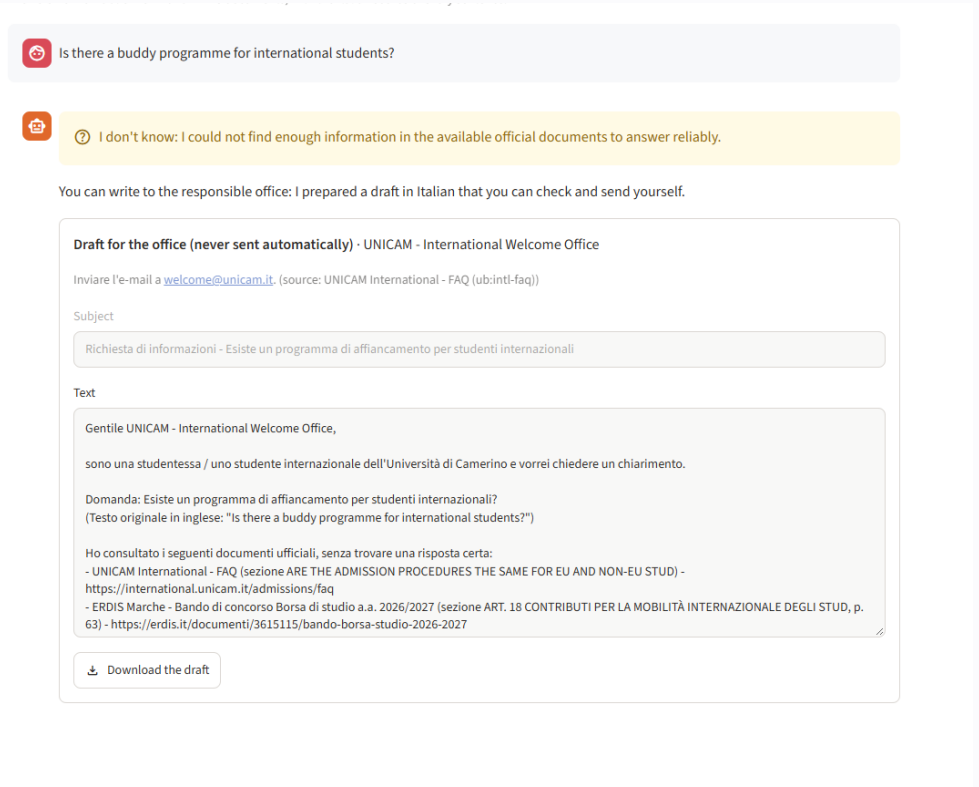
A question about football or a gym: "I don't know", no e-mail.

Permissions

A visitor never gets an answer built from a student-only notice.

In the demo



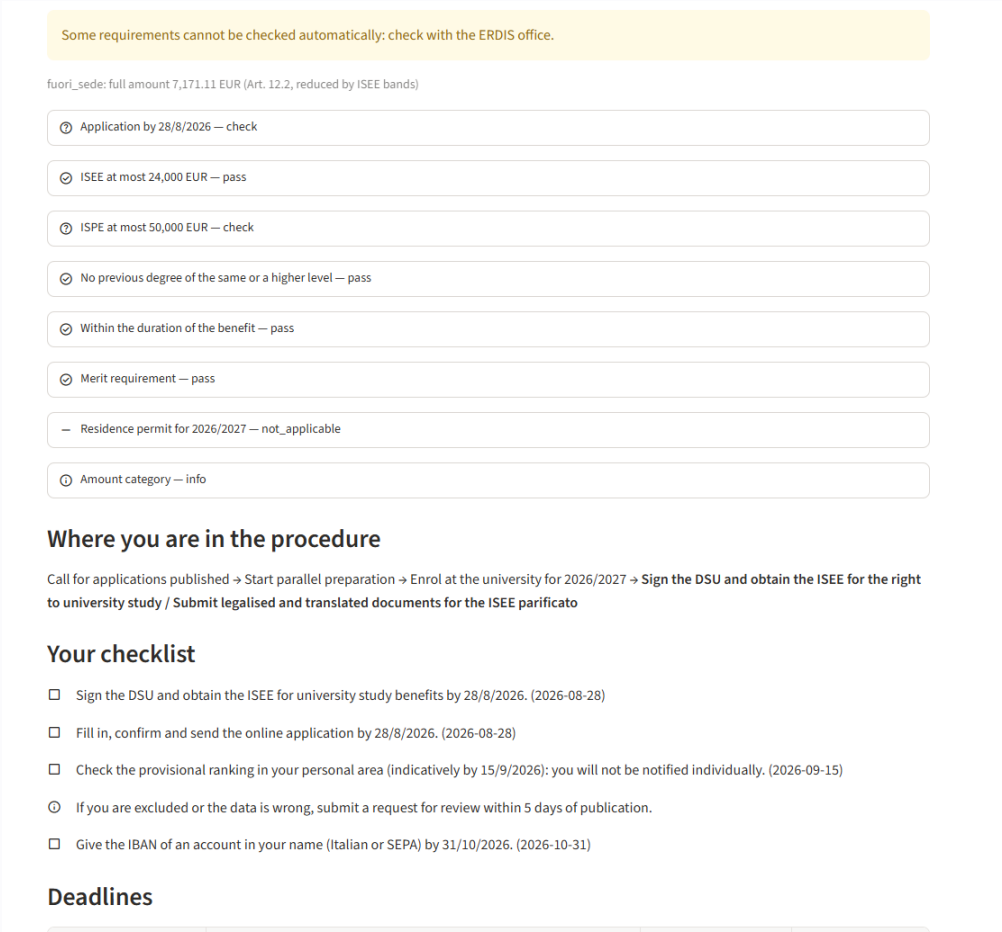


Left: a French question, answered in French with the Italian source. Right: a visitor asks about a student-only programme and gets "I don't know" with a draft for the international office.

60 sec

FROM QUESTION TO ANSWER

Scholarship guide



`No language model`

Credits by year, ISEE and ISPE limits, previous degree, residence permit and deadlines are checked by code, each rule with its article.

`Where you are`

The steps done and the next ones, from the student's facts and the date.

`15 of 15 test scenarios correct`

Fictional students with the checklist written from the call.

`Nothing stored`

Credits and ISEE stay in the request; never logged, never sent to the language model.

EVIDENCE

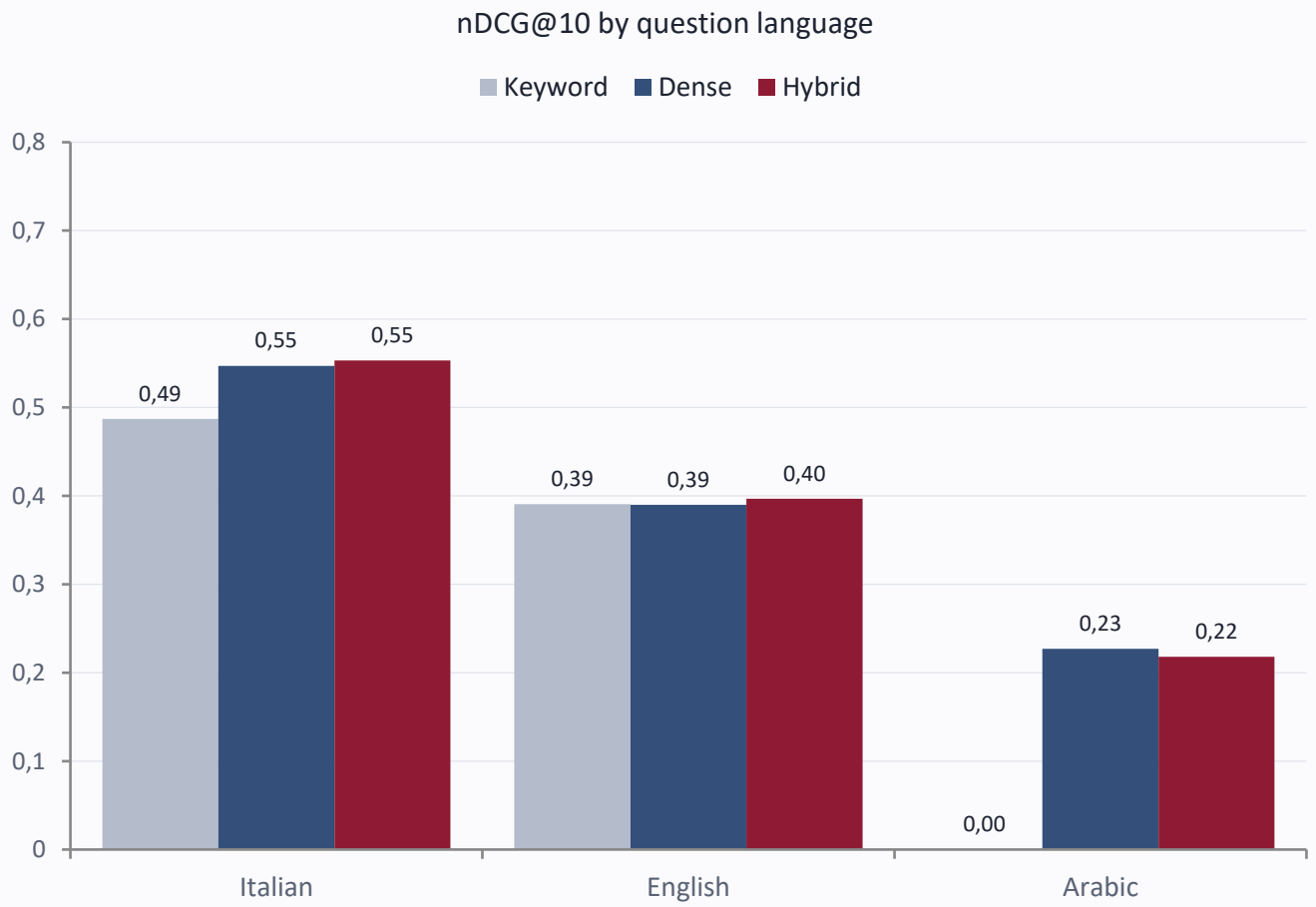
Data sources and protocol

DATA SOURCE	INTENDED USE	REMAINING GAP
42 official documents	Corpus: 2,330 passages with section, page and offsets	One university and one call; a snapshot
Official FAQ questions (42 Italian)	Questions with known answers: the FAQ answer and the article it cites	Few questions per topic
Translations: 30 English, 30 Arabic, 616 in 14 more languages	The same questions in every language	Machine translations; native review pending
6 synthetic notes (labelled)	Test the roles: student-only and office-only content	Not real notices

EVALUATION PROTOCOL

Tune on one split · freeze the test split with a SHA-256 hash · measure once · compare systems with paired randomization tests.

Keyword search fails on Arabic



100%

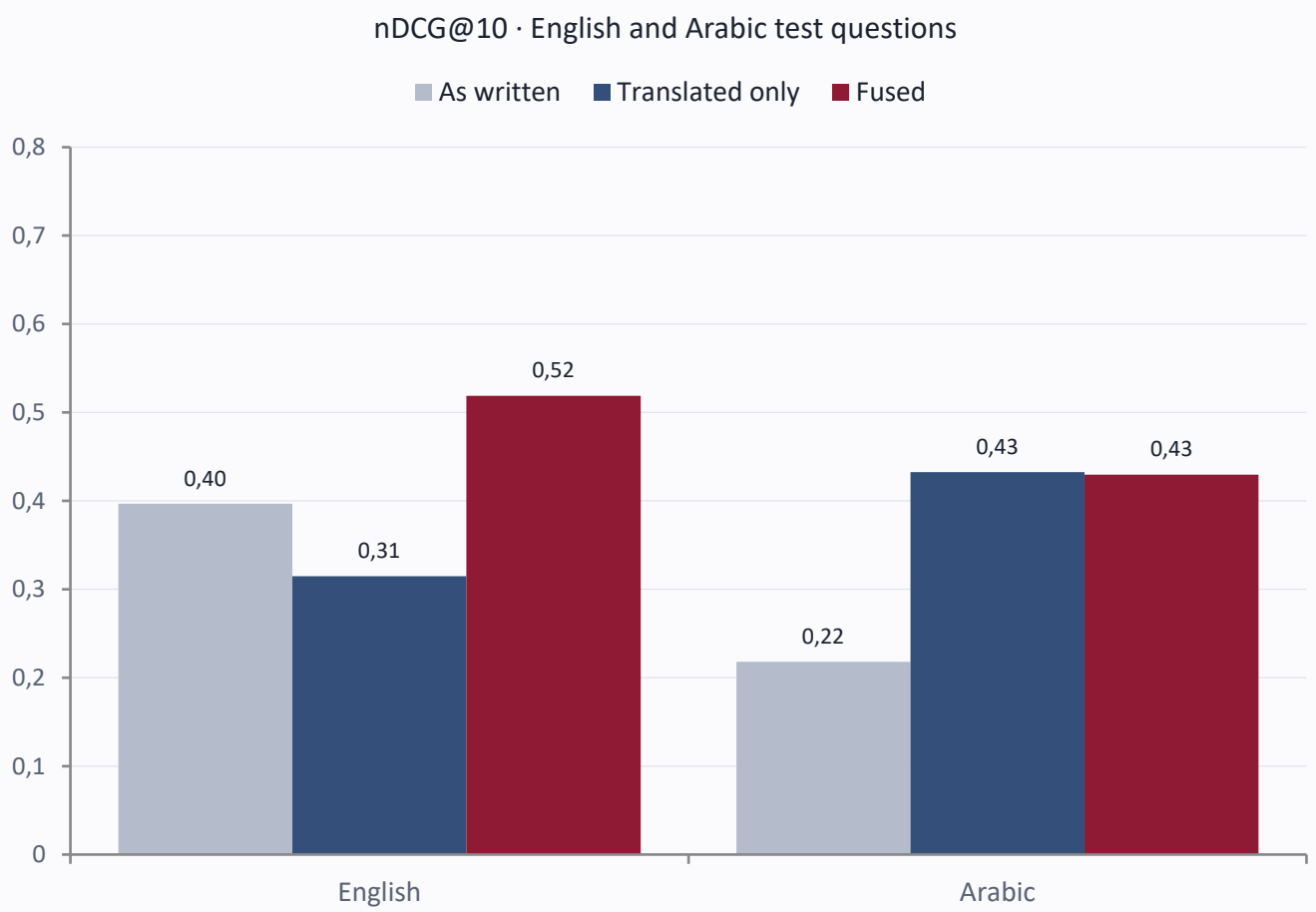
of the 26 Arabic test questions: keyword search finds no relevant passage in the top 10.

Permissions

0 leaks in 16,626 checked passages.

150 sec

Translate the question, then fuse



ALL 102 QUESTIONS

0.41 → 0.51

nDCG@10 with fusion · p = 0.0012

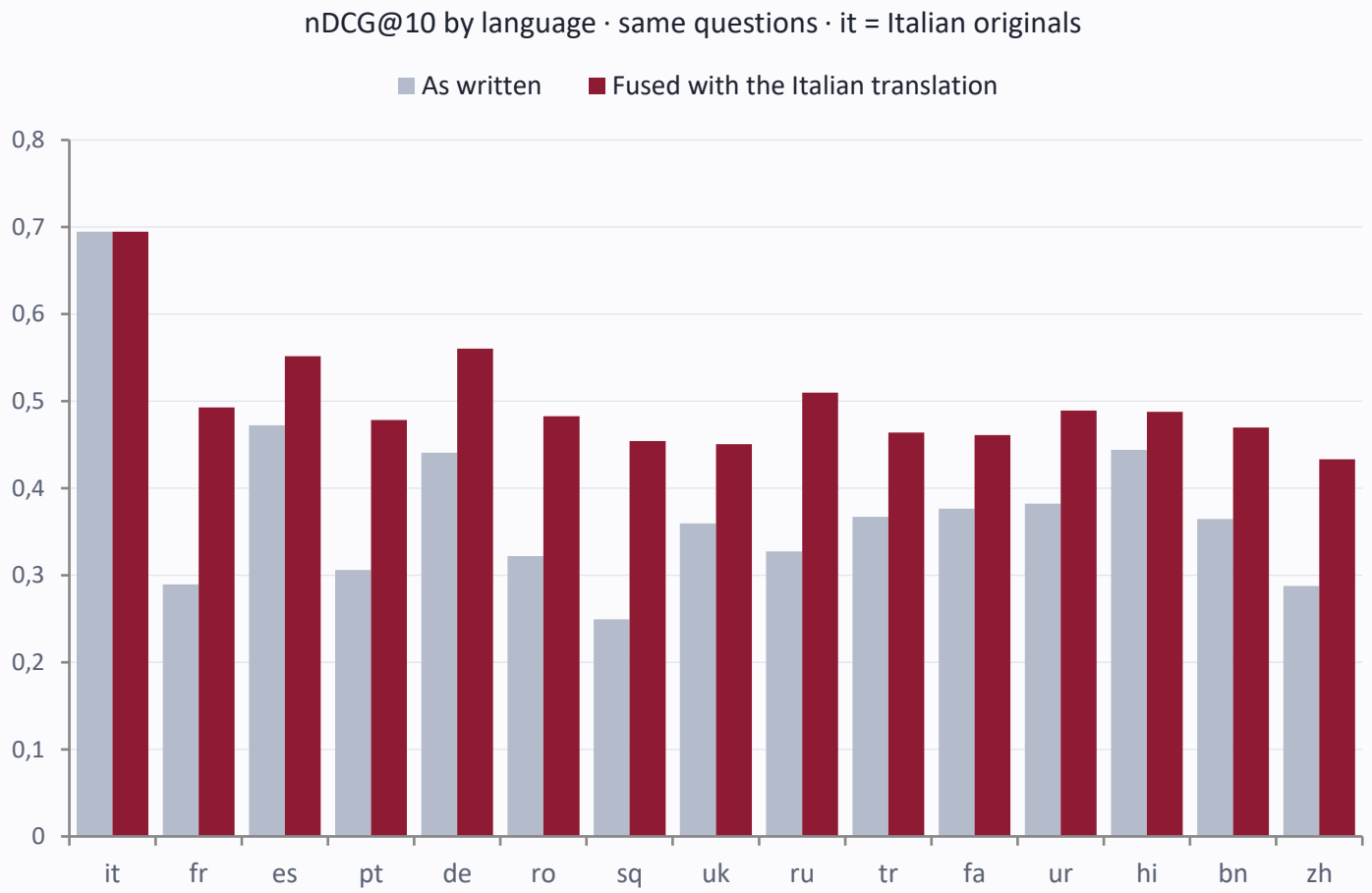
English

Translation alone is worse: English questions find the English version of the call, which the ground truth rarely marks. Measured, not yet confirmed.

150 sec

EVIDENCE

17 languages



99.8%

questions whose language is detected correctly

99.5%

answers written in the language of the question

WITHOUT FUSION

"Insufficient evidence" for 74% of in-scope questions; 11% with fusion (p = 0.0001).

150 sec

Answers: extractive, model, model + verifier

102 TEST QUESTIONS	EXTRACTIVE (NO MODEL)	MODEL, NO VERIFIER	MODEL + VERIFIER
Answered (not "I don't know")	69 of 102	87 of 102	**86 of 102**
Citation precision	0.41	0.78	**0.78**
Sentences without a valid citation	0%	7%	**0%**
Answer in the language of the question	93%	100%	**100%**
"I don't know" on out-of-scope questions	100%	100%	**100%**
Cost per answer (USD)	0	0.00023	**0.00032**
Latency, median (ms)	28	2,111	**2,976**

The verifier's check uses the same model, so it is not an independent measure of faithfulness; a human review of 30 answers is the independent check (pending). In the 14 added languages citation precision is 0.87.

150 sec

07

QUESTIONS

OUTCOME

What is done and what comes next

01 · Assistant

Answers in 17 languages, each sentence with its Italian source.

02 · Guide

Eligibility, checklist and calendar from fixed rules.

03 · Evidence

Frozen test sets, results per language, reproducible numbers.

LIMITS

Non-Italian questions are machine translations; the ground truth misses some relevant passages; the verifier sometimes removes too much; one university, one call.

NEXT STEPS

Questions written by real international students and checked by native speakers; human review of answers; more documents after licence checks; per-language "I don't know" thresholds.

Use of AI tools: a coding assistant for code and documents; a language model for translations and answers (0.41 USD for the whole project). Declared in the report.

DISCUSSION

Questions?

Demo, report and code ready to show on Friday.

Ahmad Kassar · Complex Systems Design

30 sec + Q&A